

DPS912 Lab 6: Server/Client Communications**Due: Thursday, July 9, 2026 (11:59pm)**

In this lab you will create two processes: **server** and **client**. They will communicate with each other through a socket connection. The relationship of the **server** to the **client** is that of a master to a slave. The **server** will send commands to the **client** and the **client** will obey and respond. This type of communications is called synchronous, in that the **server** does not send any more data until it has received a response from the **client**.

- The **server** and **client** will communicate via a socket file **/tmp/lab6_your_initial**.
- The socket type is **AF_UNIX, SOCK_STREAM**. The socket family is **AF_UNIX**.
- The **client** will run inside an infinite while loop whereas the **server** will send specific commands then quit.
- The sequence of commands are as follows:
 1. The **server** will send the command "**Pid**" to the **client** to request the client's pid.
 2. The **client** will respond with "**This client has pid <pid>**", where <pid> can be obtained from **getpid()**.
 3. The **server** will print out the response from the **client**.
 4. The **server** will then send the command "**Sleep**" to the **client** to request the **client** to sleep for 5 seconds.
 5. The **client** will sleep for 5 seconds then respond to the **server** with "**Done**".
 6. When the **server** receives "**Done**" from the **client**, the **server** will send the command "**Quit**" to tell the **client** to shutdown.
 7. The **client** will break out of its infinite while loop, close the socket connection, then return 0.
 8. The **server** will close all socket connections, unlink the socket file **/tmp/lab6_youe_initial**, then return 0.
- Be sure to add error handling in your code.
- Please add debug printf's/cout's throughout your code to keep track of what your **server** and **client** are doing.

- A sample run for the **server** and **client** are as follows (be sure to run the **server** first):

```
miguelwatler@ubuntu:~/src/Labs/Lab5$ ./server
Waiting for the client...
client connected to the server
server: accept()
The server requests the client's pid
server: This client has pid 2248
The server requests the client to sleep
The server requests the client to quit
miguelwatler@ubuntu:~/src/Labs/Lab5$
```

```
miguelwatler@ubuntu:~/src/Labs/Lab5$
miguelwatler@ubuntu:~/src/Labs/Lab5$
miguelwatler@ubuntu:~/src/Labs/Lab5$
miguelwatler@ubuntu:~/src/Labs/Lab5$ ./client
A request for the client's pid has been received
This client is going to sleep for 5 seconds
This client is quitting
miguelwatler@ubuntu:~/src/Labs/Lab5$
```

Research Questions:

- 1) Based on your experience in this lab, explain one advantage of using an AF_UNIX socket (like /tmp/lab6) instead of an AF_INET/TCP socket for server↔client communication on the same machine. Explain one disadvantage.
- 2) In this lab, the server waits for a reply before sending the next command. Describe one benefit of this synchronous design. Describe one situation where asynchronous (non-blocking or multi-threaded) communication would be better.

Assignment Submission:

- Complete all steps, Add all output-screenshot and explanations (if required) to a MS-Word file.
- Add the following declaration at the top of MSWORD file and [source code](#)

```

/*****
* DPS912-Lab6
* I declare that this lab is my own work in accordance with Seneca Academic Policy.
* No part of this assignment has been copied manually or electronically from any other source
* (including web sites) or distributed to other students.
*
* Name: _____ Student ID: _____ Date: _____
*
*
*****/

```

- Please answer the following two declarations:

- **D1)** On a scale from 1 to 5, **How much did you use generative AI to complete this assignment?**
 - where:
 - **1** means you did not use generative AI at all
 - **2** means you used it very minimally
 - **3** means you used it moderately
 - **4** means you used it significantly
 - **5** means you relied on it almost entirely
 - **Your answer :**

- **D2)** On a scale from 1 to 5, **How confident are you in your understanding of the generative AI support you utilized in this assignment, and in your ability to explain it if questioned?**
 - where:
 - **1** means "Not confident at all – I do not understand the generative AI support I used and cannot explain it."
 - **2** means "Slightly confident – I understand a little, but I have many uncertainties."
 - **3** means "Moderately confident – I understand the majority of the support, though some parts are unclear."
 - **4** means "Very confident – I understand most of the AI support well and can explain it with minor gaps."
 - **5** means "Extremely confident – I fully understand the generative AI support I used and can clearly explain or justify it if asked."
 - **Your answer :**
- Please submit the Source code (zip all .c, .h, and makeFiles)

Important Note:

- **LATE SUBMISSIONS for labs.** There is a deduction of 10% for Late assignment submissions, and after three days it will grade of zero (0).
- This labs should be submitted along with a video-recording which contains a detailed walkthrough of solution. Without recording, the assignment can get a maximum of 1/3 of the total.
 - Note: In case you are running out of time to record the video, you can submit the assignment (source code + screenshots) by the deadline and submit the video within 24 hours after the deadline.